Dipanshu Raj

Machine Learning Engineer

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Professional Summary

B.Tech Computer Science student at BIT Mesra with strong foundations in Python, C, DSA, Linux, and Git. Focused on Machine Learning and Computer Vision, with hands-on experience implementing ML algorithms from scratch using NumPy.

Education

Birla Institute of Technology, Mesra, *B.Tech in Computer Science* 2024 – 2028

Loyola High School, *Intermediate (PCM)* 2023

Loyola High School, *Matriculation* 2021

Skills

Python • Machine Learning • NumPy • TensorFlow • Data Visualization • C • Linux • Data Structures & Algorithms • Git

Projects

Neural Network (MLP) From Scratch (Numpy) 2026 – 09/2026
Implemented a fully connected Multi-Layer Perceptron using only NumPy with forward propagation, backpropagation, ReLU, Softmax, cross-entropy loss, and mini-batch gradient descent; trained for 26-class handwritten letter recognition.

Logistic Regression from Scratch (NumPy)
Built binary & multiclass logistic regression with Softmax and Cross-Entropy; achieved 95.13% test accuracy.

Linear Regression from Scratch (NumPy)
Implemented gradient descent, MSE cost function, vectorized training, and performance visualizations.

Certifications

Machine Learning

Mathematics for Machine Learning & Data Science

CS50P: Introduction to Programming with Python

CS50x: Introduction to Computer Science

Languages

- Hindi
- English